Normalization case study 2

Database Management Systems (The University of the South Pacific)

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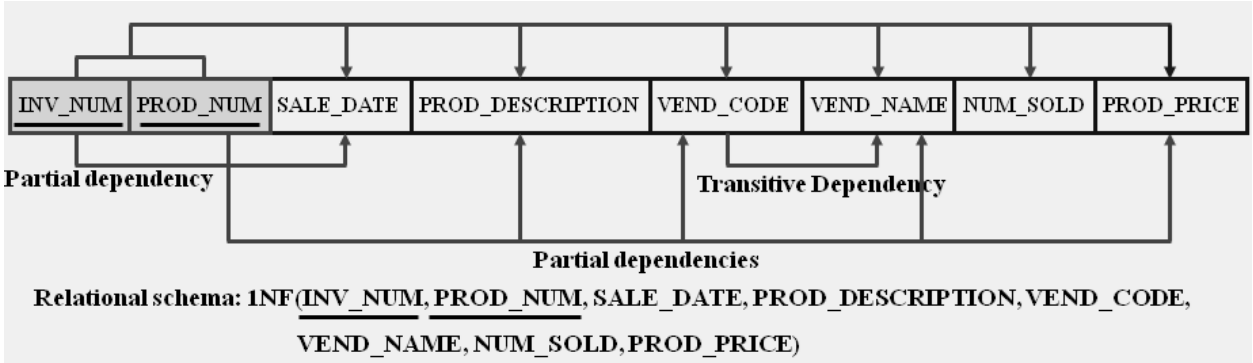
Consider the INVOICE table structure with sample invoice records below:

TABLE P6.3					
ATTRIBUTE NAME	SAMPLE VALUE	SAMPLE VALUE	SAMPLE VALUE	SAMPLE VALUE	SAMPLE VALUE
INV_NUM	211347	211347	211347	211348	211349
PROD_NUM	AA-E3422QW	QD-300932X	RU-995748G	AA-E3422QW	GH-778345P
SALE_DATE	15-Jan-2016	15-Jan-2016	15-Jan-2016	15-Jan-2016	16-Jan-2016
PROD_LABEL	Rotary sander	0.25-in. drill bit	Band saw	Rotary sander	Power drill
VEND_CODE	211	211	309	211	157
VEND_NAME	NeverFail, Inc.	NeverFail, Inc.	BeGood, Inc.	NeverFail, Inc.	ToughGo, Inc.
QUANT_SOLD	1	8	1	2	1
PROD_PRICE	$49.95	$3.45	$39.99	$49.95	$87.75

1. Write the relational schema, draw its dependency diagram and identify all dependencies, including all partial and transitive dependencies. Identify the normal forms for each table structure you created

   Assume:
   a. the table does not contain repeating groups
   b. any invoice number may reference more than one product.



Hint: This table uses a composite primary key.

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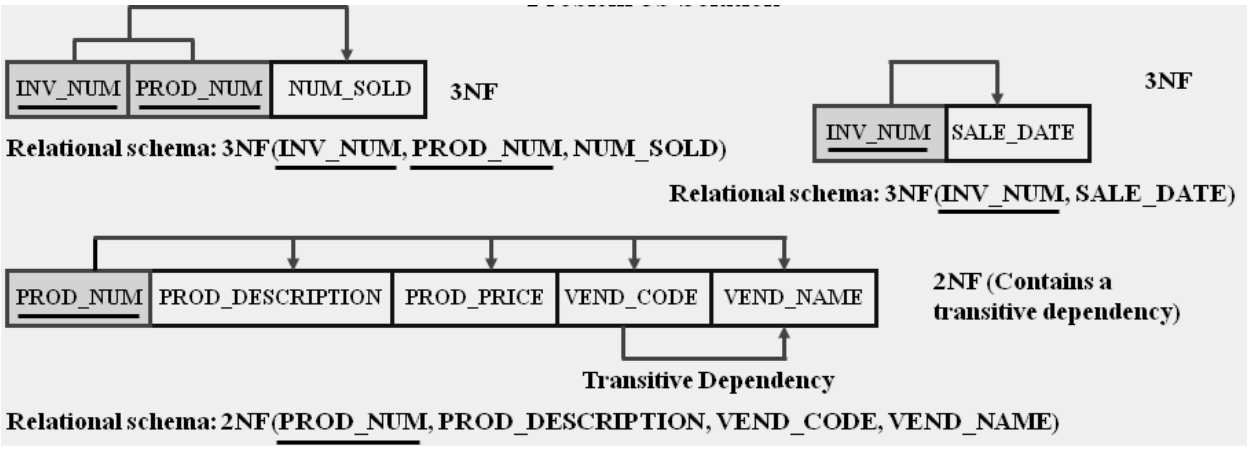
2. Remove all partial dependencies, write the relational schema, and draw the new dependency diagrams. Identify the normal forms for each table structure you created.

   Assume:
   a. any given product is supplied by a single vendor
   b. a vendor can supply many products

   From a&b, we can conclude the following dependency exists:
   PROD_NUM → PROD_DESCRIPTION, PROD_PRICE, VEND_CODE, VEND_NAME

   Hint: Your actions should produce three dependency diagrams



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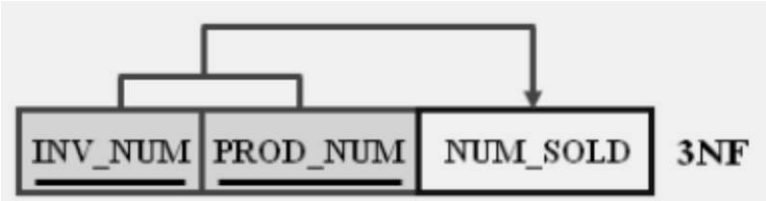
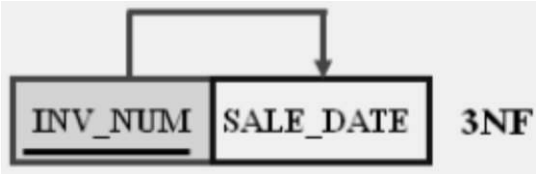
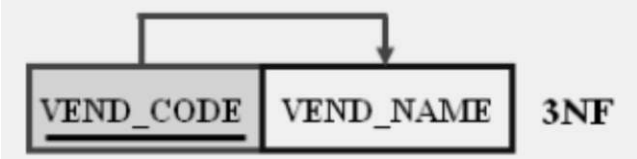
3. Remove all transitive dependencies, write the relational schema, and draw the new dependency diagrams. Identify the normal forms for each table structure you created.



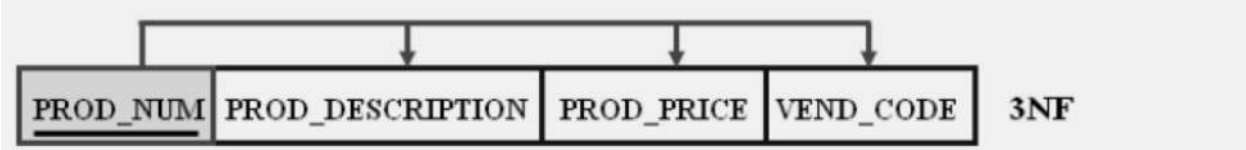
Relational schema: 3NF(INV_NUM, PROD_NUM, NUM_SOLD)



Relational schema: 3NF(INV_NUM, SALE_DATE)



Relational schema: 3NF(VEND_CODE, VEND_NAME)



Relational schema: 3NF(PROD_NUM, PROD_DESCRIPTION, VEND_CODE)

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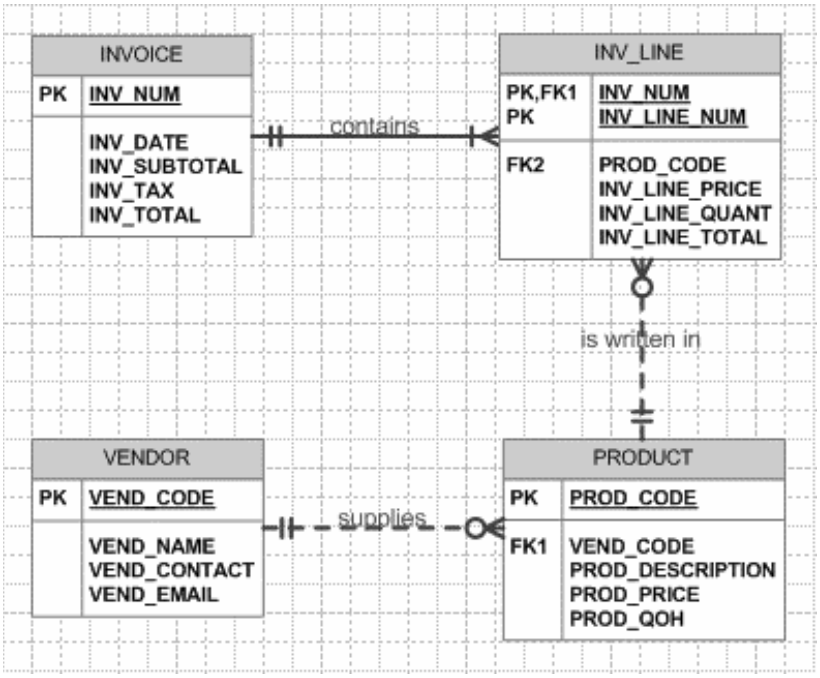
4. Draw the Crow's Foot ERD.

Note:

a. The dependency diagrams cannot show the nature (1:1, 1:M, M:N) of the relationships, hence the ER Diagrams remain crucial to the design effort.
b. Complex design is impossible to produce successfully without some form of modeling (e.g, ER modeling, or some other modeling methodology).
c. As the preceding decompositions demonstrate, the dependency diagrams are a valuable addition to the designer's toolbox - Normalization is likely to suggest the existence of entities that may not have been considered during the modeling process.
d. If information or transaction management issues require the existence of attributes that create other than 3NF or BCNF conditions, the proper dependency diagrams will at least force awareness of these conditions.

The invoicing ERD, accompanied by its relational diagram, is shown below. (The relational diagram only includes the critical PK and FK components, plus a few sample attributes, for space considerations.)



Crow's Foot Invoicing ERD

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